

Assignment (AI)

A working **AI Study Assistant**:

- RAG over documents
- Tool-using agent (calculator + doc search)
- FastAPI backend
- Streamlit UI
- Deployable project + GitHub repo



WEEK 1 — Core LLM + RAG Foundation

◆ Day 1: Python + API Basics

Focus

- Requests, JSON, file handling



Task

Build CLI tool:

- Input topic
- Fetch Wikipedia summary
- Format output nicely



Demo Target

✓ Run CLI:

```
Enter topic: Python
→ Returns structured Wikipedia summary (clean formatted text)
```

◆ Day 2: Prompt Engineering Basics

Focus

- Prompt templates
- Output formatting



Task

Build "Prompt Playground CLI":

- 5 prompt types:

- summarize
- explain simply
- extract bullets
- quiz generator
- compare topics

Demo Target

✓ Same input produces 5 different LLM-style outputs:

```
Input: Machine Learning
→ 5 structured responses displayed in terminal
```

◆ Day 3: Embeddings + Chunking

Focus

- Embeddings concept
- Text chunking

Task

- Load 2–3 text files
- Split into chunks
- Generate embeddings (mock or real)
- Store in memory list / FAISS

Demo Target

✓ Print:

```
Query: "What is gradient descent?"
→ Top 3 most relevant chunks displayed
```

◆ Day 4: Vector Search (Mini RAG v0)

Focus

- Similarity search

Task

- Use FAISS / Chroma
- Retrieve relevant chunks from documents

Demo Target

✓ Terminal demo:

Ask: "Explain backpropagation"
→ Shows retrieved document chunks ranked by relevance

◆ Day 5: RAG Pipeline (First full system)

Focus

- Retrieval + LLM response generation

Task

Build full RAG:

- Load docs
- Retrieve chunks
- Send to LLM prompt

Demo Target

✓ Working Q&A:

Question: "Summarize Chapter 2"
→ Final answer generated from document context

◆ Day 6: Improve RAG Quality

Focus

- chunk size tuning
- prompt improvement
- citations

Task

- Add source snippets in response
- Improve prompt grounding

Demo Target

✓ Answer includes sources:

Answer: ...

Sources:

- chunk 1
- chunk 3

◆ Day 7: Week 1 Integration Demo

Focus

- stability + cleanup

Task

- unify RAG pipeline
- handle edge cases

Demo Target (IMPORTANT MILESTONE)

✓ End-to-end demo:

Upload text → Ask question → Get grounded answer with sources

 This is your **first “real product demo” moment**

WEEK 2 — Agents + App + Deployment

◆ Day 8: Tools Concept

Focus

- tools = functions for LLM

Task

Create 2 tools:

- calculator
- document search tool

Demo Target

✓ Manual tool execution:

```
Input: 5 * 12  
→ Calculator tool returns 60
```

```
Input: "Explain AI"  
→ RAG tool returns context
```

◆ Day 9: Basic Agent Logic

Focus

- routing decisions

Task

- Build rule-based agent:
 - math → calculator
 - theory → RAG
 - else → LLM

Demo Target

✓ Agent decides correctly:

```
Input: "What is 25*8?"  
→ Uses calculator tool automatically
```

```
Input: "Explain transformers"  
→ Uses RAG tool
```

◆ Day 10: LangChain-style Agent (optional upgrade)

Focus

- tool calling abstraction

Task

- Replace rules with LLM decision layer (if ready)

Demo Target

✓ LLM chooses tool dynamically

◆ Day 11: FastAPI Backend

Focus

- APIs

Task

Create:

- `/chat`
- `/upload`
- `/ask`

Demo Target

✓ Postman / curl demo:

```
POST /ask → returns RAG response JSON
```

◆ Day 12: Streamlit UI

Focus

- simple frontend

Task

- chat UI
- file upload UI

Demo Target

✓ Live UI:

- upload file
 - ask question
 - get response in browser
-

◆ Day 13: Polish + Stability

Focus

- error handling
- logging
- memory (optional)

Task

- fix failures

- improve UX

Demo Target

✓ Smooth flow:

Upload → Ask → Response → No crashes

◆ Day 14: Final Capstone Demo Day

Focus

- packaging + showcase

Task

- finalize repo
- README
- architecture diagram

FINAL DEMO TARGET

Full system demo:

1. Open Streamlit UI
2. Upload document
3. Ask:
 - "Explain this in simple words"
 - "Give key points"
 - "Solve a related math problem"
4. System:
 - uses RAG
 - uses calculator tool when needed
 - gives grounded answers

Outcome after 2 weeks

The fresher will have built:

- A working **RAG system**
- A basic **AI agent**
- A full-stack **LLM application**

- Real debugging experience with LLM behavior
- A portfolio-ready project